

2D Derivation Target for the 2022 Magnetic-Field-Inspired Navigation Paper

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Purpose

This note fixes a concrete 2D target for the clean Python implementation. The 2022 paper is written in a general 3D vector-field form, but the current benchmark platform is planar. The goal of this document is to define a precise 2D reduction target for:

- the robot motion direction l_a ,
- the obstacle surface normal n_o ,
- the artificial surface current l_o ,
- the perpendicular / opposite current $l_o^\perp$,
- the boundary-following field F_b ,
- the collision-avoidance field F_a .

This document is not yet a proof that the current code matches the paper. It is the derivation target that the code should match.

Paper Context

The 2022 paper states:

- the robot has a local sensor of radius r_s ,
- the robot velocity direction is l_a ,
- the closest obstacle point is described by vector r_o ,
- the obstacle normal is taken opposite to r_o ,
- the artificial current l_o is the projection of l_a onto the local obstacle surface,
- F_b is induced by l_o ,
- F_a is induced by $l_o^\perp$,
- both F_b and F_a should be orthogonal to robot motion and therefore not change robot speed.

2D Setup

We work in $\mathbb{R}^2$. Let:

$$p \in \mathbb{R}^2 \quad (1)$$

denote the robot position,

$$v = \dot{p} \in \mathbb{R}^2 \quad (2)$$

denote the robot velocity, and

$$r_o \in \mathbb{R}^2 \quad (3)$$

denote the vector from the robot to the currently selected closest obstacle point.
The robot speed is

$$\|v\| \quad (4)$$

and the robot motion direction is

$$l_a = \begin{cases} \frac{v}{\|v\|}, & \|v\| > 0 \\ 0, & \|v\| = 0 \end{cases} \quad (5)$$

Local Surface Normal and Tangent

Following the paper assumption, the obstacle surface normal at the currently sensed closest point is taken opposite to r_o :

$$n_o = -\frac{r_o}{\|r_o\|} \quad (6)$$

whenever $\|r_o\| > 0$.

The orthogonal projector onto the local obstacle tangent line is therefore

$$P_t = I - n_o n_o^\top \quad (7)$$

Artificial Surface Current l_o

The paper defines l_o as the projection of the robot motion direction l_a onto the local obstacle surface.

In 2D, the natural reduction is:

$$\tilde{l}_o = P_t l_a = l_a - (l_a^\top n_o) n_o \quad (8)$$

If $\|\tilde{l}_o\| \geq \epsilon$, the unit artificial current is

$$l_o = \frac{\tilde{l}_o}{\|\tilde{l}_o\|} \quad (9)$$

If $\|\tilde{l}_o\| < \epsilon$, the paper says to use the unit vector of the previously proposed current. Therefore the 2D target rule is:

$$l_o = \begin{cases} \frac{\tilde{l}_o}{\|\tilde{l}_o\|}, & \|\tilde{l}_o\| \geq \epsilon \\ l_{o,\text{prev}}, & \|\tilde{l}_o\| < \epsilon \end{cases} \quad (10)$$

2D Cross-Product Reduction

To reduce the paper's 3D cross-product expressions into a planar form, define the 90-degree rotation matrix

$$J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad (11)$$

and define the scalar 2D cross product

$$a \times_2 b = a_x b_y - a_y b_x \quad (12)$$

for $a, b \in \mathbb{R}^2$.

If $a, b \in \mathbb{R}^2$ are embedded into $\mathbb{R}^3$ with zero third component, then:

$$a \times b = (0, 0, a \times_2 b) \quad (13)$$

and

$$a \times (0, 0, \beta) = -\beta J a \quad (14)$$

These identities are the basis of the exact 2D reduction target below.

2D Target for F_b

The paper boundary-following field follows the charged-particle style magnetic construction:

$$B_b = \frac{l_o \times v}{\|r_o\|} \quad (15)$$

$$F_b = c l_a \times B_b \quad (16)$$

Reducing this to 2D gives the target:

$$B_b^{(z)} = \frac{l_o \times_2 v}{\|r_o\|} \quad (17)$$

$$F_b^{2D} = -c B_b^{(z)} J l_a \quad (18)$$

or equivalently,

$$F_b^{2D} = -c \frac{l_o \times_2 v}{\|r_o\|} J l_a \quad (19)$$

Important property

Because Jl_a is orthogonal to l_a ,

$$(F_b^{2D})^\top l_a = 0 \quad (20)$$

which is the 2D target expression for the paper statement that F_b does not change robot speed.

2D Target for F_a

After checking the old ROS `field=2` branch more carefully, the best 2D target for the current clean implementation is not a second magnetic cross-product law. The stronger implementation evidence is:

- F_b is the magnetic boundary-following term,
- F_a is a short-range radial collision term,
- F_a is projected to remain orthogonal to the motion direction so it does not change speed.

Define the raw short-range repulsion

$$\tilde{F}_a = -c_\perp \left(\frac{1}{\|r_o\|} - \frac{1}{r_{la}} \right) \frac{r_o}{\|r_o\|} \quad (21)$$

for $\|r_o\| \leq r_{la}$. The working 2D target is then the orthogonal projection of this repulsion onto the subspace perpendicular to l_a :

$$F_a^{2D} = \left(I - l_a l_a^\top \right) \tilde{F}_a \quad (22)$$

or equivalently,

$$F_a^{2D} = \tilde{F}_a - \left(\tilde{F}_a^\top l_a \right) l_a \quad (23)$$

Important property

By construction,

$$(F_a^{2D})^\top l_a = 0 \quad (24)$$

which is the 2D target expression for the paper statement that F_a does not change robot speed.

Thresholded Activation Target

In implementation, the paper uses:

- F_b only when $\|r_o\| \leq r_l$
- F_a only when $\|r_o\| \leq r_{la}$

So the 2D paper target is:

$$F_b^{\text{active}} = \begin{cases} F_b^{2D}, & \|r_o\| \leq r_l \\ 0, & \text{otherwise} \end{cases} \quad (25)$$

$$F_a^{\text{active}} = \begin{cases} F_a^{2D}, & \|r_o\| \leq r_{la} \\ 0, & \text{otherwise} \end{cases} \quad (26)$$

Local Sensing Target

The paper assumes:

- only currently visible local sensor points are available
- the closest point is chosen from those readings
- if several close readings fall within threshold δr , averaging is used for the special non-unique closest-point case

Therefore the 2D implementation target is:

1. collect only sensor-visible obstacle points within radius r_s ,
2. select the minimum-distance sensed vector,
3. form the averaged vector over readings with distance less than $d_{\min} + \delta r$,
4. use the averaged vector only when the paper Section 4.4 condition is satisfied.

What This Means For The Next Implementation Step

The next implementation step should treat the following as the exact 2D target:

- $n_o = -r_o / \|r_o\|$
- $l_o = \text{normalize}((I - n_o n_o^\top) l_a)$, with the ϵ -memory rule
- $F_b^{2D} = -c \frac{l_o \times_2 v}{\|r_o\|} J l_a$
- $F_a^{2D} = (I - l_a l_a^\top) \tilde{F}_a$, where $\tilde{F}_a = -c_\perp \left(\frac{1}{\|r_o\|} - \frac{1}{r_{la}} \right) \frac{r_o}{\|r_o\|}$

This target may still need revision after checking more carefully against the paper equations. However, it now matches the old ROS implementation evidence for the short-range collision term much better than the rejected magnetic $l_o^\perp$ -based target.